

BRAIN TUMOR CLASSIFICATION: A DEEP LEARNING APPROACH

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ABSTRACT

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1 INTRODUCTION

Brain tumors are a prevalent disease among the elderly, this is concerning since in Canada: “the population aged 85 and older has doubled since 2001”(Hallman et al., 2022). As a result, this will likely lead to increased cases of brain tumors. The fast-paced nature of this disease makes early detection vital for a patient’s survival and diagnosis is typically done by expert analysis of medical images (i.e., MRI scans). This task lends itself to deep learning since a model can be trained to detect brain tumors in images based on already classified images by experts. This model can be used to analyze brain scans to verify the expert’s diagnosis or in circumstances where an expert is not readily accessible. This model can be a vital tool in the upcoming years as the number of cases of brain tumors is expected to increase. Thus the project proposed is a deep learning model capable of detecting tumors in brain scans, it takes as an input a brain scan in JPG format and it will classify the scan as tumorous or not tumorous. For tumorous scans, it will be capable of classifying the tumor from the pituitary, meningioma, and glioma types.

2 BACKGROUND AND RELATED WORK

1. **Automated Brain Tumor Detection and Classification Using Convolutional Neural Networks:** This is an IEEE study which demonstrates an automated method for diagnosing and classifying brain tumors using Convolutional Neural Networks (CNNs) as they are exceptionally good at processing high-resolution medical images (M. Nagraju, Apr. 2024). The researchers used high-resolution MRI scans of the brain taken from several different datasets which emphasize the importance of quality images when it comes to training the model to achieve near-perfect accuracy. When it comes to the architecture, the researchers integrated attention mechanisms as well as multi-scale feature fusion to increase the model’s sensitivity to different tumor features. The model achieves a 99.76% accuracy in the identification as well as classification of tumors in MRI scans, surpassing the tumor classification experts.
2. **Deep Learning Suite for 3D Image Segmentation** DeepMedic is an open-source deep learning framework developed to process medical images, especially brain lesion segmentation. It uses a 3D deep convolutional neural network coupled with a 3D fully connected conditional field (deepmedic.org, 2024). This allows for the processing of 3D medical images such as MRI and CT scans. This model has implemented multi-scale input architecture, thus it can process images at different scales, allowing it to detect features varying in size, improving its accuracy.

3. **Enhancing brain tumor detection in MRI images through explainable AI using Grad-CAM with Resnet 50** This study by BMC Medical Imaging used Resnet 50 which is a CNN architecture known for its ability to extract small details from an image (M. M. Musthafa, May 2024). To increase the transparency of the model, making the tumor identification process more interpretable, the model uses Grad-CAM (Gradient-weighted Class Activation Mapping) which illustrates which part of the image the model focuses on to make its decisions. Data augmentation was also used which is the process of modifying the images in already existing datasets to make the model generalize and learn better. The model achieved an accuracy of 98.52% as well as a high F1 score.
4. **A robust grey wolf-based deep learning for brain tumour detection in MR images** This study employed contrast enhancement and skull stripping to remove the excess noise, thus allowing the model to focus on the important features of the MRI scans (A. Geetha, 2018). They also use the Fuzzy C-Means Clustering (FCM) algorithm to isolate the regions of interest for tumor detection. Furthermore the Gray Co-occurrence Matrix (GLCM) and Gray-Level Run-Length Matrix (GRLM) enable the model to extract specific features from the MRI scans. This study takes a unique approach of using the Deep Belief Network (DBN), optimized using Grey Wolf Optimization (GWO) to identify brain tumors as opposed to CNNs used by the other studies mentioned thus far.
5. **Brain Tumor Detection Based on Deep Learning Approaches and Magnetic Resonance Imaging** This study fine-tuned a YOLOv7 deep learning model using the transfer learning technique to improve its performance. In addition, feature enhancements such as Convolutional Block Attention Module (CBAM) and Spatial Pyramid Pooling Fast+ (SPPF+) were added (A. B. Abdusalomov, 2023). The study concluded that the model had a 99.5% accuracy in detecting glioma, meningioma, and pituitary brain tumors.

3 DATA PROCESSING

The project utilizes two main datasets, a 3064 image dataset from figshare and a 3000 image dataset from Kaggle. The figshare dataset contains T1-weighted contrast-enhanced brain tumor images collected between 2005 and 2010 from Nanfang Hospital, Guangzhou, China, and General Hospital, Tianjin Medical University, China from 233 patients (Cheng, 2017). Furthermore, the dataset organizes the images in the 3 tumor types: the pituitary with 930 samples, glioma with 1426 samples, and meningioma with 708 samples. An additional dataset from Kaggle, the Br35H dataset, will be used, which contains 1,500 tumorous and non-tumorous images each, however, only the non-tumorous images were used since the tumorous were not classified into the 3 aforementioned tumor types (Hamada, 2021).

One of the preprocessing challenges was varying image sizes. This was resolved via a script that zero-padded images to the correct aspect ratio and then resized them to 224x224. Furthermore, each dataset stored the images in different formats: One of the datasets provided the images in a Matlab struct variable and the other stored the images as JPG files. To ensure consistency, all of the images were converted to JPG format.

Due to limited resources available on the web, our dataset wasn't large enough, which posed challenges for effective model training and generalization. Furthermore, our dataset exhibited a significant class imbalance, with the Meningioma class and Pituitary class being underrepresented. To tackle these issues, we performed augmentation on the training data using the transform library from PyTorch and performed the following transformations:

1. **Rotations:** Rotates the image by a random degree between 0 and 90 degrees, adding variety in orientation and helping the model generalize across different rotations.
2. **Horizontal Flip:** Flips the image horizontally with a probability of 100%, simulating the scenarios where tumors may appear on the opposite side, thus enhancing the model's robustness.
3. **Vertical Flip:** Similar to the horizontal flip, this transformation flips the image vertically with a probability of 100%, providing further positional variability for the model.

4. **Random Affine:** Applies random affine transformations such as small translations (up to 10% of the image size). This transformation slightly shifts the image content, which aids the model in generalizing to minor positional variations of tumors.
5. **Color Jitter:** Adjusts the brightness, contrast, saturation, and hue of the images randomly, allowing the model to become more resilient to variations in lighting and color conditions.
6. **Random Blur:** Applies a random Gaussian blur to the image with a kernel size of 5 and a sigma in the range (2,5). This mimics scenarios where the image is slightly out-of-focus, ensuring that the model can still detect tumors even if the image is not sharply focused.

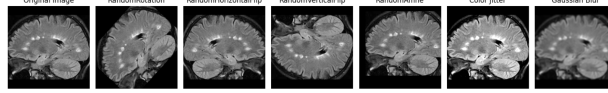


Figure 1: Example Transoformations of A Selected Image.

To address the class imbalance, we applied specific augmentations which helped ensure that the model generalizes well across all categories while avoiding excessive oversampling. For underrepresented classes such as the Meningioma class and the Pituitary class, we applied targeted transformations to increase variability and prevent bias. On the other hand, we limited augmentation for dominant classes such as the Non-tumorous class and the Glioma class to maintain the overall distribution. To provide a clearer overview of the augmentation strategies for each class, the details are summarized in the following table:

Class	Type	Transformations Applied
Non-tumorous	Underrepresented	Applied all 6 transformations
Meningioma	Dominant	Randomly applied 2 transformations to each image
Glioma	Dominant	Randomly applied 3 transformations to each image
Pituitary	Underrepresented	Randomly applied 5 transformations to each image

Table 1: Summary of Data Augmentation by Class

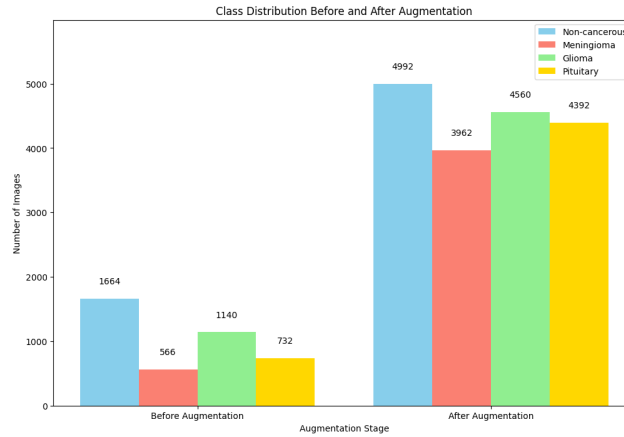


Figure 2: Class Distribution Before and After Augmentation.

4 BASELINE MODEL

Support Vector Machine (SVM) is a machine-learning algorithm that allows the computer to classify any input that is given to it (Wahl, 2017). SVM was used as the classifier for the baseline model as it is suitable for processing complex, high-dimensional data and it is an effective classifier for detecting patterns in MRI scans with minimal preprocessing.

SVM operates by “finding the optimal hyperplane that maximizes the margin between the closest data points of opposite classes” (IBM, 2023). In n -dimensional space, multiple hyperplanes can exist that differentiate between the classes, which is why the margin between the data points is maximized to find the optimal hyperplane that represents the best decision boundary between the classes (IBM, 2023). As seen in Figure 1, the lines parallel to the optimal hyperplane are called support vectors and they cross the closest points to the decision boundary, thus they determine the position and orientation of the optimal hyperplane.

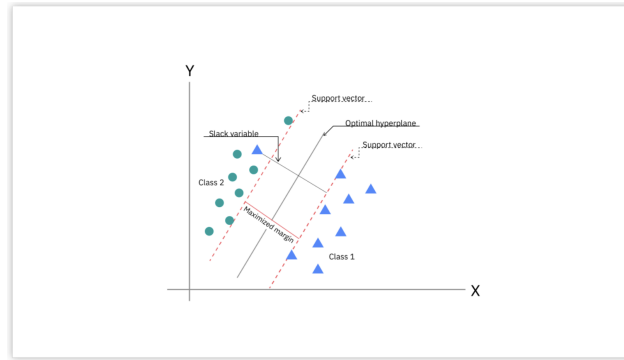


Figure 3: Optimal Hyperplane and Support Vector (IBM, 2023).

The baseline model employs the SVM with a linear kernel, which generates a linear decision boundary in the feature maps. The C parameter in SVM is used to control the trade-offs between minimizing the training error and maximizing the margin. The higher the C parameter, the more emphasis is placed on minimizing the training error, potentially at the expense of smaller margins (<https://www.baeldung.com/cs/ml-svm-c-parameter>). In the baseline model, the SVM is initialized with a C value of 1.0 which maximizes the penalty for misclassifications. This configuration is needed for medical diagnosis where minimizing the false negatives is crucial to ensure patients with a brain tumor are not misclassified.

Instead of extracting individual features from the input images using convolutional layers or pre-trained models like AlexNet, the raw images were input directly into the SVM classifier. As a result, the model relies solely on the classifier to learn patterns from the pixel values in the input MRI scans.

During the training of the baseline model, one of the challenges faced was the model using too much memory. The SVM does not support batch training, thus we were unable to divide the dataset into smaller batches, which could have been done efficiently using data loaders. As a result, the entire dataset had to be loaded into the memory causing the RAM capacity to be exceeded and ultimately crash. To fix this issue, we utilized both CPU and GPU memory for storing and processing the training images. After resetting the runtime, we were able to store all images in memory without exceeding the limits during training. This allowed the model to fully train without running into memory limitations.

As a result of a large number of batches and a relatively large image size, training the SVM with the MRI scans presented a significant runtime challenge. Using the T4 GPU on Google Colab, training the model would take roughly three hours. The T4 GPUs work better for general tasks with strong support for machine learning, however they need to catch up when it comes to fast processing for deep learning tasks compared to the newer GPUs. The L4 GPU offers optimized memory bandwidth, faster processing and data handling for deep learning tasks. This improved the training efficiency thus drastically reducing the training time.

5 PRIMARY MODEL

The primary model makes use of Transfer learning. It takes the input image (size 224 x 224) and sends it to the feature extractor of Alexnet. Alexnet produces very efficient feature maps that capture critical features of the input image. It consists of 5 convolutional layers. The following are the specs of Alexnet’s feature extractor (depicted in Figure 4):

- **Convolutional layer 1:** 96 kernels, size 11x11, stride length of 4
- **Overlapping max pooling layer:** 3x3 kernel size, stride length of 2
- **Convolutional layer 2:** 256 kernels, size 5x5, padding of 2
- **Overlapping max pooling layer:** 3x3 kernel size, stride length of 2
- **Convolutional layer 3:** 384 kernels, size 3x3, padding of 1
- **Convolutional layer 4:** 384 kernels, size 3x3, padding of 1
- **Convolutional layer 5:** 256 kernels, size 3x3, padding of 1
- **Overlapping max pooling layer:** 3x3 kernel size, stride length of 2

The main issue faced was that the feature maps produced by Alexnet took a long time to extract since the dataset is quite large, the extraction process as a whole took even longer. The output of Alexnet's feature map extractor is of size $256 \times 6 \times 6$. This is sent to a fully connected layer with 64 neurons. The output of this layer is sent to another fully connected layer with four neurons for classification to indicate whether it was a meningioma, glioma, pituitary, or non-tumorous image. The ReLU (Rectified Linear Unit) is used as the activation function between the two layers. The network itself has 590,148 total trainable parameters $((256 \times 6 \times 6 \times 64 + 64) + (64 \times 4 + 4))$.

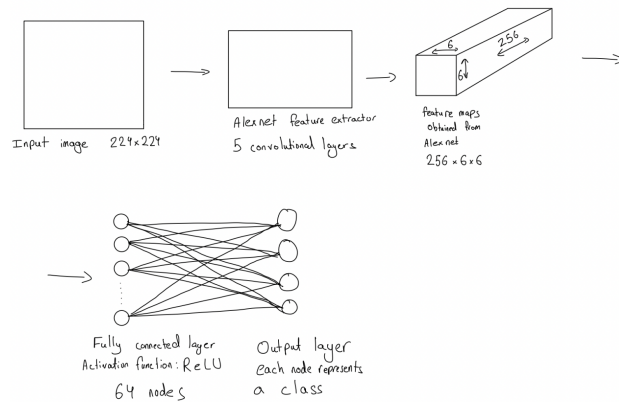


Figure 4: Illustration of the Model Architecture.

6 RESULTS

A new dataset used for testing data was employed to ensure the quality of the models are accurately reported. The dataset is a 1311-image dataset from Kaggle, it consists of 405 non-tumorous images, 306 meningioma images, 300 glioma images and 300 pituitary images (Santhosh, 2024). The following sections are the results of the baseline and primary model on the new training dataset.

6.1 BASELINE RESULTS

The baseline model's performance during training and evaluation was assessed using accuracy. On validation data, it had an accuracy of 99.98%, showing signs of overfitting to the current validation images.

From the confusion matrix below in Figure 5, we can see the model's performance in all four categories:

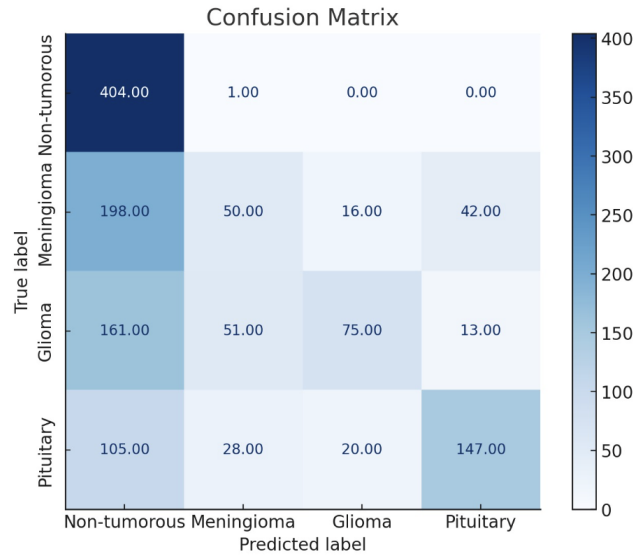


Figure 5: Baseline Model Confusion Matrix.

Class	Precision	Recall	F1 Score
Non-tumorous	46.54%	100%	63.52%
Meningioma	38.46%	18.94%	25.38%
Glioma	67.57%	26.13%	37.69%
Pituitary	72.77%	49.00%	58.57%

Table 2: Performance Metrics by Class

The confusion matrix illustrates a strong performance in identifying the Non-tumorous class, with almost all cases correctly identified, resulting in a 100% recall. This highlights the baseline model's ability to distinguish between healthy and tumorous brain scans. The model also performs relatively well in the Pituitary class, with a precision of 72.77% and a recall of 49%. This demonstrates that the model maintains a balance between over-predicting and under-predicting, as evidenced by the 58.75% F1 score.

However, the model struggles with identifying the Meningioma and Glioma classes. For Meningioma, the model has a very low recall of 18.94% and a precision of 38.46%, which shows severe misclassification of Meningioma cases into other classes. This indicates overlapping features with other classes, along with poor distinctiveness in the data for this tumor type. Similarly, the Glioma class achieves a precision of 67.57% and a low recall of 26.13%, also demonstrating misclassification of Glioma cases into other classes which is further echoed by the poor F1 score of 37.69%. Overall, the poor precision, recall, and F1 scores suggest significant room for improvement in the model.

6.2 PRIMARY MODEL RESULTS

In evaluating our model, various metrics were used to assess its performance. The model's performance during training and validation was assessed using accuracy and loss metrics. As seen in Figure 6, the training accuracy steadily improved from 84% at Epoch 1 to approximately 100% at Epoch 30 while the validation accuracy stabilized around 97% after slight fluctuations at the start. Furthermore, our training loss decreased consistently, while validation loss remained low but showed occasional fluctuations.

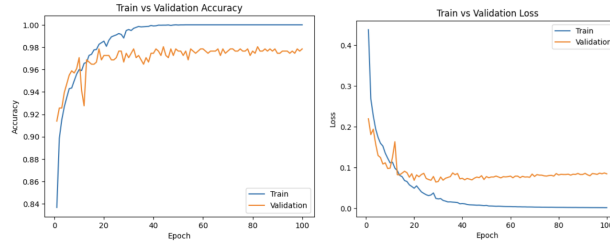


Figure 6: Train and Validation Graphs.

From the confusion matrix in Figure 7, we can assess the model's performance across all 4 categories:

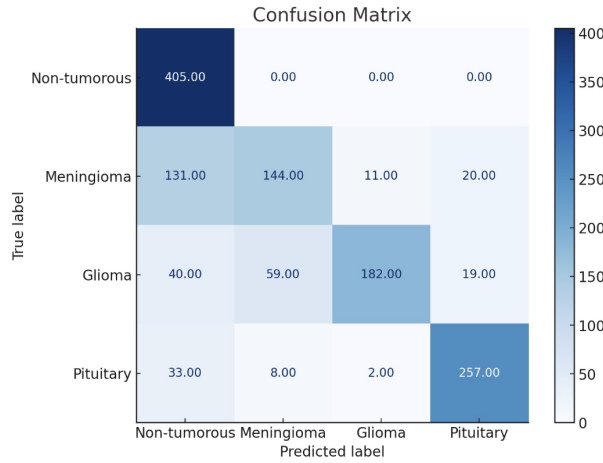


Figure 7: Primary Model Confusion Matrix

Class	Precision	Recall	F1 Score
Non-tumorous	66.5%	100%	79.9%
Meningioma	68.2%	47.0%	55.7%
Glioma	93.3%	60.1%	73.5%
Pituitary	86.8%	85.7%	86.2%

Table 3: Performance Metrics by Class

The confusion matrix highlights strong performance in identifying the Non-tumorous and Pituitary classes with 405 and 257 correct predictions respectively. These results highlight the model's strength in identifying cases with more distinct and less ambiguous features. For non-tumorous samples, all cases were predicted correctly demonstrating that the model is adept at distinguishing a healthy brain from a tumorous brain, Pituitary tumors also exhibit a similar result.

However, the model exhibits challenges with Meningioma detection. Although 144 classes are correctly identified, 131 Meningioma cases are misclassified as Non-tumorous. This indicates that the model is often confusing the Meningioma class with the Non-tumorous class. Conversely, misclassification of Meningioma as Glioma or Pituitary is much less frequent. This confusion may arise because the dataset contains 405 Non-tumorous samples compared to 306 Meningioma samples, creating a class imbalance that may potentially bias the model towards the dominant Non-tumorous class. Additionally, some of the features present in the Meningioma samples may overlap with the features in the Non-tumorous class making it harder for the model to distinguish between the two

classes. Similarly, the Glioma class also suffers from this issue: the model predicts 182 of these samples correctly, 59 samples as Meningioma, and 40 as Non-tumorous.

As previously mentioned meningioma and non-tumorous were commonly confused by the model. Figure 8 depicts random samples of the two classes and what the model predicted for each:

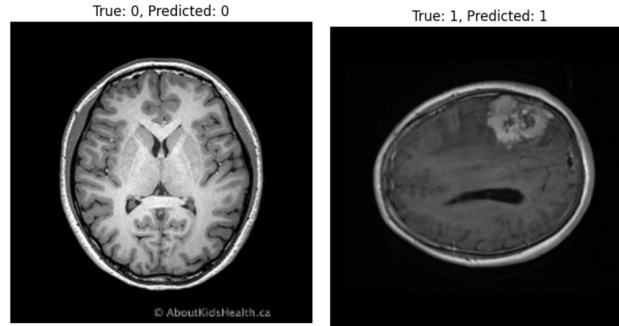


Figure 8: Depiction of two brain samples, the left image is Non-tumorous (Class 0), and the right is Meningioma (Class 1).

7 ETHICAL CONSIDERATIONS

Since the model will be trained using medical images this causes a privacy concern since these images tend to contain sensitive information about patients. Furthermore, a lack of diversity in the data set can be problematic since the same type of brain tumor could look different in two patients with different characteristics such as race, ethnicity, and gender. Having a diverse dataset therefore ensures the model is fair and accurate for people of all backgrounds and traits. There are many considerations in cases where the model produces inaccurate results. A false positive, where a non-tumorous patient is misclassified as having a specific tumor type, may lead to unnecessary stress and tests. A false negative, where a tumor is missed, can delay critical treatment, potentially resulting in severe outcomes. Moreover, misclassifying one type of tumor as another could also lead to incorrect treatment plans.

8 PROJECT DIFFICULTY/QUALITY

The baseline model achieved an F1 score of 46.29% thus exhibiting the inherent difficulty of the project as the baseline model employs an SVM which is well suited for classification applications. The F1 score of the primary model was 73.825%. This drastic improvement shows that the final model is a good starting point and a stronger choice than traditional machine learning methods. However, for sensitive medical applications such as this, the margin of error is slim. Thus this report concludes that the model requires further revision.

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